



September 2017 Examination

EC400

Probability and Statistical Inference - Solutions

Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

You are not allowed to use an electronic calculator

1. **(7 points - 10 minutes)** Suppose that 1 in 10,000 people (0.01%) has a particular disease. If a person has the disease, the test will give a positive result with a probability of 0.99. If a person does not have the disease, the test will give a negative result with a probability of 0.99. Find the probability of not having the disease if the test comes out positive. Provide your answer as a simplified fraction.

$$\text{Answer: } \Pr(\text{no disease} \mid + \text{ test}) = P[B|A] = \frac{P[B] \cdot P[A|B]}{P[A]}$$

$$\frac{\Pr(\text{no disease}) \cdot \Pr(+ \text{ test} \mid \text{no disease})}{\Pr(+ \text{ test} \mid \text{disease}) \cdot \Pr(\text{disease}) + \Pr(+ \text{ test} \mid \text{no disease}) \cdot \Pr(\text{no disease})} = \frac{\frac{9999}{100} \cdot \frac{1}{10000}}{\frac{99}{100} \cdot \frac{1}{10000} + \frac{1}{100} \cdot \frac{9999}{10000}} = \frac{9999}{10098} = \frac{101}{102}$$

(no need to give the latter fraction)

2. **(8 points - 10 minutes)** Let X be a continuous random variable with pdf $f_X(x) = \frac{c}{x^2}$ if $1 < x < \infty$ and $f_X(x) = 0$ otherwise.

- (a) Determine the value of the constant c .

$$\int_1^\infty f(x) dx = \int_1^\infty \frac{c}{x^2} dx = -cx^{-1} \Big|_1^\infty = c \equiv 1, \text{ so } c = 1$$

- (b) Consider the continuous random variable $Y = e^{\sqrt{X}}$. What is the pdf of Y and what is the probability of $Y > 0$?

We have a one-to-one relation between Y and X , and clearly $e < y < \infty$.

Using the change of variable method with $u(x) = e^{\sqrt{x}}$ and $v(y) = (\ln y)^2$, with $v'(y) = 2(\ln y) \frac{1}{y}$, so that

$$\begin{aligned} g(y) &= f(v(y)) \cdot 2(\ln y) \frac{1}{y}, \text{ for } e < y < \infty, 0 \text{ otherwise} \\ &= \frac{1}{(\ln y)^4} 2(\ln y) \frac{1}{y} = \frac{2}{y(\ln y)^3} \text{ for } e < y < \infty, 0 \text{ otherwise} \end{aligned}$$

$\Pr(Y > 0) = 1$ as our random variable can only take values in excess of e .

3. **(18 points - 20 minutes)** Let X have an exponential distribution with parameter $\lambda > 0$. Its density is given by

$$f(x) = \frac{1}{\lambda} \exp\left(-\frac{1}{\lambda}x\right), 0 \leq x < \infty, 0 \text{ otherwise,}$$

and $E(X) = \lambda$. Let $X_1, \dots, X_n$ be an iid sample drawn from this distribution.

- (a) Derive the maximum likelihood estimator for λ ($\hat{\lambda}_{MLE}$).

By independence, the joint density is given by

$$f(x_1, \dots, x_n) = \frac{1}{\lambda^n} \exp\left(-\frac{1}{\lambda} \sum_{i=1}^n x_i\right) \equiv L(\lambda)$$

From here obtain the log-likelihood

$$\ln L(\lambda) = -n \ln(\lambda) - \frac{1}{\lambda} \sum_{i=1}^n x_i$$

and by solving the FOC, we obtain our MLE estimator:

$$-\frac{n}{\hat{\lambda}} + \frac{1}{\hat{\lambda}^2} \sum_{i=1}^n x_i = 0 \Rightarrow \hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i$$

The MLE estimator therefore is given by $\bar{X}$.

- (b) *The variance of X is given by λ^2 . Discuss how you can obtain a consistent estimator for the variance. Explain your claim clearly.*

We simply use $\hat{\lambda}^2 = (\bar{X})^2$. By Slutsky, we have $\text{plim } \hat{\lambda}^2 = \text{plim } ((\bar{X})^2) = (\text{plim } \bar{X})^2$. Khinchine LLN ensures $\text{plim } \bar{X} = E(X_i) = \lambda$. Therefore we obtain $\text{plim } \hat{\lambda}^2 = \lambda^2$ and consistency is established. Obviously they can suggest another estimator and show its consistency as well

- (c) *Provide a method of moment estimator for the variance of X . Explain the intuition underlying your proposal.*

We know $\lambda^2 = E(X^2) - (E(X))^2$, by replacing the population moments by their sample analogue we obtain $\hat{\lambda}_{MM}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2 \equiv \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. (This estimator really makes use of two moments $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$ without imposing any special relation between the mean and the variance as is the case here).

Because of the special relation between the mean and the variance, in fact it would also be correct to use the moment $E\left(X^2 - \frac{\lambda^2}{\text{variance}} - \frac{\lambda^2}{E(X)^2}\right) = 0$, so that

$$E(X^2 - 2\lambda^2) = 0 \text{ and use } \frac{1}{2} \left(\frac{1}{n} \sum_{i=1}^n X_i^2 \right).$$

4. **(12 points - 15 minutes)** Let $f(x, y) = 3/2$, $x^2 \leq y < 1$, $0 \leq x \leq 1$, be the joint pdf of X and Y .

- (a) *Find the conditional distribution of Y given $X = x$.*

We need to use the result that $f(y|x) = \frac{f(x,y)}{f(x)}$.

Obtain $f(x) = \int_{x^2}^1 (3/2) dy = \frac{3}{2} (1 - x^2)$ with $0 \leq x \leq 1$, 0 otherwise.

Therefore $f(y|x) = \frac{1}{1-x^2}$ for $x^2 \leq y < 1$ and $0 \leq x \leq 1$

- (b) Are X and Y independent.

Since $f(y|x)$ depends on x , we know that X and Y are dependent!

- (c) Find $\Pr(X \geq 1/2, Y \geq 1/2)$. You may want to use a sketch when answering this question.

Because of the boundaries involving y , we should observe

$$\begin{aligned} \Pr(X \geq 1/2, Y \geq 1/2) &= \Pr(X \geq 1/2, Y \geq 1/4) - \Pr(X \geq 1/2, 1/4 \leq Y < 1/2) \\ &= \int_{1/2}^1 \int_{x^2}^1 3/2 dy dx - \int_{1/2}^{1/\sqrt{2}} \int_{x^2}^{1/2} 3/2 dy dx \\ &= \int_{1/2}^1 \left(\frac{3}{2} (1 - x^2) \right) dx - \int_{1/2}^{1/\sqrt{2}} \left(\frac{3}{2} \left(\frac{1}{2} - x^2 \right) \right) dx = \frac{5}{8} - \frac{1}{4}\sqrt{2}. \end{aligned}$$

5. (25 points - 30 minutes) Let us assume that the continuous random variable X "Height of an adult living in the UK" has a normal distribution with means μ_X and variance $\sigma_X^2 = 0.025$. Say a random sample of 1000 observations of heights of people in the UK was obtained, and the sample mean is given by $\bar{x} = 1.769$.

- (a) Test the hypothesis that $H_0 : \mu_X = 1.77$ against the alternative $H_A : \mu_X \neq 1.77$ using a significance level of 5%.

We need to use the Z test, given by $Z = \frac{\bar{X} - 1.77}{\sqrt{\sigma_X^2/n}} \sim N(0, 1)$ under H_0 . Given

the 5% level of significance we should reject if $\left| \frac{\bar{x} - 1.77}{\sqrt{\sigma_X^2/n}} \right| > 1.96$. Here $\frac{\bar{x} - 1.77}{\sqrt{\sigma_X^2/n}} = \frac{-0.001}{\sqrt{\frac{25}{1000000}}} = \frac{-\frac{1}{1000}}{\frac{5}{1000}} = -\frac{1}{5}$. So clearly we should not reject.

- (b) Explain how to compute the p -value for the two-sided test. Draw a sketch to illustrate. (There is no need to provide an exact numerical answer here or in the rest of the question).

Provide sketch. Compute $\Pr(|Z| > |-\frac{1}{5}|) = 2 \Pr(Z > \frac{1}{5}) = 2 * .4207 = 0.8414$

- (c) Discuss how you would obtain the power of your test in (a) if the true mean adults' height is 1.765.

We need to compute

$$\begin{aligned} \Pr\left(\left|\frac{\bar{X} - 1.77}{\sqrt{\sigma_X^2/n}}\right| > 1.96 \mid \mu = 1.765\right) &= \Pr\left(\frac{\bar{X} - 1.765}{\sqrt{\sigma_X^2/n}} > 1.96 - \left(\frac{1.765 - 1.77}{\sqrt{\sigma_X^2/n}}\right)\right) + \Pr\left(\frac{\bar{X} - 1.765}{\sqrt{\sigma_X^2/n}} < -1.96 - \left(\frac{1.765 - 1.77}{\sqrt{\sigma_X^2/n}}\right)\right) \\ &= \Pr\left(Z > 1.96 + \frac{.005}{.005}\right) + \Pr\left(Z < -1.96 + \frac{.005}{.005}\right) = \Pr(Z > 2.96) + \Pr(Z < -0.96) = .0015 + .1685 = 0.17 \end{aligned}$$

- (d) Say, in (a) we had tested the hypothesis that $H_0 : \mu_X = 1.77$ against the alternative $H_A : \mu_X > 1.77$. What can you then say about power of your test if the true mean adults' height is 1.765? Discuss this finding (compare it to the significance level). You may want to draw a sketch to illustrate.

Now we need to compute

$$\begin{aligned} & \Pr\left(\frac{\bar{X}-1.77}{\sqrt{\sigma_X^2/n}} > 1.645 \mid \mu = 1.765\right) \\ &= \Pr\left(\frac{\bar{X}-1.765}{\sqrt{\sigma_X^2/n}} > 1.645 - \left(\frac{1.765-1.77}{\sqrt{\sigma_X^2/n}}\right)\right) = \Pr(Z > 2.645) = 0.004 \end{aligned}$$

We discover then that the power of the test (our ability to reject when the null is false) is smaller than the size of the test (the probability of rejecting the null when it is true). This is a problem that arises due to us not recognizing the possibility that the true size could (and indeed is) smaller than 1.77 in this case. (Sketch optional)

6. (18 points - 20 minutes) Let $X_1, \dots, X_n$ an iid sample drawn from the distribution $X \sim N(0, \sigma^2)$ where σ^2 is unknown but finite.

- (a) Drawing on the standard estimator for the sample variance, someone proposes to use the following estimator for σ^2 :

$$\tilde{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n X_i^2.$$

Is this estimator unbiased and consistent? Proof your claims. Discuss the difference between these two concepts.

The estimator is biased: $E(\tilde{\sigma}^2) = \frac{1}{n-1} \sum_{i=1}^n E(X_i^2) = \frac{n}{n-1} \sigma^2$ as $E(X_i^2) = E(X) = 0$ $Var(X_i) = \sigma^2$.

The estimator is consistent $\text{plim } \tilde{\sigma}^2 = \text{plim } \frac{n}{n-1} \cdot \text{plim } \frac{1}{n} \sum_{i=1}^n X_i^2 = 1 \cdot E(X_i^2)$ using Khinchine LLN and recognizing that $\text{plim } \frac{n}{n-1} = \lim \frac{n}{n-1} = 1$.

Unbiasedness is a finite sample property (over repeated sampling is your estimator on average correct) whereas consistency is a large sample property (when the sample is large can you get close to the truth).

- (b) Instead of providing an estimate of σ^2 for a given sample, let $n = 10$, you are asked to construct a 95% confidence interval. You may continue assuming that μ is known and equal to 0. What details do you need to compute the confidence interval and what information does the interval provide?

Hint: You may recall $\frac{1}{\sigma^2} \sum_{i=1}^n X_i^2 \sim \chi_n^2$.

The 95% confidence interval provides an interval $[c_L(s_X^2), c_U(s_X^2)]$ that contains the true parameter with 95% confidence, i.e., $\Pr(c_L(s_X^2) \leq \sigma^2 \leq c_U(s_X^2)) = 0.95$.

We use $\Pr(\chi_{n,.975}^2 \leq \frac{1}{\sigma^2} \sum_{i=1}^n X_i^2 \leq \chi_{n,.025}^2) = \Pr\left(\frac{1}{\chi_{n,.975}^2} \geq \frac{\sigma^2}{\sum_{i=1}^n X_i^2} \geq \frac{1}{\chi_{n,.025}^2}\right) = \Pr\left(\frac{\sum_{i=1}^n X_i^2}{\chi_{n,.975}^2} \geq \sigma^2 \geq \frac{\sum_{i=1}^n X_i^2}{\chi_{n,.025}^2}\right)$ and our interval therefore is $\left[\frac{\sum_{i=1}^n x_i^2}{\chi_{n,.025}^2}, \frac{\sum_{i=1}^n X_i^2}{\chi_{n,.975}^2}\right]$, where $\Pr(\chi_n^2 < \chi_{n,.0.975}^2) = 0.025$ and $\Pr(\chi_n^2 > \chi_{n,.0.025}^2) = 0.025$.

7. (12 points - 15 minutes) Let $f(x)$ and $F(x)$ denote, respectively, the pdf and the cdf of the random variable X .

(a) Define the conditional pdf of X given $X > x_0$ where x_0 a fixed number. This function, also called the Hazard function, finds application in a problem of the time until death, given survival until time x_0 and show that $f(x|X > x_0)$ is a pdf.

$f(x|X > x_0) = \frac{f(x)}{\Pr(X > x_0)} = \frac{f(x)}{1-F(x_0)}$. It a pdf as it is non-negative, and $\int f(x|X > x_0)dx = \int_{x_0}^{\infty} \frac{f(x)}{1-F(x_0)}dx = \frac{1}{1-F(x_0)} \int_{x_0}^{\infty} f(x) dx = \frac{1-F(x_0)}{1-F(x_0)} = 1$

(b) Let $f(x) = e^{-x}$, $0 < x < \infty$, zero elsewhere. Compute $\Pr(X > 2|X > 1)$. (You are not expected to provide an exact numerical answer)

Observe $F(x) = \int_0^x e^{-t}dt = -e^{-t}]_0^x = 1 - e^{-x}$, so $f(x|X > 1) = \frac{e^{-x}}{e^{-1}} = e^{1-x}$.

Then $\Pr(X > 2|X > 1) = \int_2^{\infty} e^{1-t}dt = -e^{1-t}]_2^{\infty} = e^{-1}$. This result is consistent with memorylessness of the exponential distribution.

In fact students could also have used Bayes' to state $\Pr(X > 2|X > 1) = \frac{\Pr(X > 2 \cap X > 1)}{\Pr(X > 1)}$. Since $X > 2$ is a subset of $X > 1$ we have $\frac{\Pr(X > 2 \cap X > 1)}{\Pr(X > 1)} = \frac{\Pr(X > 2)}{\Pr(X > 1)} = \frac{e^{-2}}{e^{-1}} = e^{-1}$.